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                        A* SEARCH DRY RUN
=====

Iteration 1
-----
Current Node : A

Open List :
Empty

Closed List :
Empty

Current Values
g( A ) = 0
h( A ) = 7
f( A ) = 7

Iteration 2
-----
Current Node : B

Open List :
C f = 8

Closed List :
['A']
```

```
Current Values
g( B ) = 2
h( B ) = 6
f( B ) = 8

Iteration 3
-----
Current Node : E

Open List :
C f = 8
D f = 12

Closed List :
['A', 'B']

Current Values
g( E ) = 4
h( E ) = 2
f( E ) = 6

Iteration 4
-----
Current Node : C

Open List :
D f = 9
D f = 12
```

Closed List :
['A', 'B']

Current Values
g(E) = 4
h(E) = 2
f(E) = 6

Iteration 4

Current Node : C

Open List :
D f = 9
D f = 12

Closed List :
['A', 'B', 'E']

Current Values
g(C) = 4
h(C) = 4
f(C) = 8

Iteration 5

Current Node : D

Open List :
D f = 12

Closed List :
['A', 'B', 'E', 'C']

Current Values
g(D) = 6
h(D) = 3
f(D) = 9

Iteration 6

Current Node : G

Open List :
D f = 12

Closed List :
['A', 'B', 'E', 'C', 'D']

Current Values
g(G) = 8
h(G) = 0
f(G) = 8

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=====
FINAL RESULT
=====
Optimal Path : A -> B -> E -> D -> G
Total Path Cost : 8

Process finished with exit code 0
```

Q2:

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=====
QUESTION 2 : MINIMAX WITH ALPHA-BETA PRUNING
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```
Step 1
Current Node : MAX
Alpha = -inf
Beta  = inf

Step 2
Current Node : MIN ( B )
Alpha = -inf
Beta  = inf
```

```
Step 3
Leaf Node : 3
Alpha = -inf
Beta  = inf
Updated Beta = 3
```

```
Step 4
Leaf Node : 5
Alpha = -inf
Beta  = 3
Updated Beta = 3
```

```
Step 5
Leaf Node : 6
```

```
Alpha = -inf
Beta  = 3
Updated Beta = 3
Updated Alpha = 3
```

```
Step 6
Current Node : MIN ( C )
Alpha = 3
Beta  = inf
```

```
Step 7
Leaf Node : 9
Alpha = 3
Beta  = inf
Updated Beta = 9
```

```
Step 8
Leaf Node : 1
Alpha = 3
Beta  = 9
Updated Beta = 1
Alpha-Beta Cutoff
Updated Alpha = 3
```

```
=====
FINAL RESULT
=====
Best Move for MAX : Left Subtree
Final Minimax Value : 3
Pruned Nodes : [2]

Process finished with exit code 0
```